Chapter 4: Continuity

Section: Continuous Functions

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

From Limits to Continuity

- Section 4.1: $\lim_{x \to p} f(x) = q$: the value $f(p)$ was *irrelevant* (even undefined!)

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- Preview: compositions, a purely topological characterization via open sets, algebra of continuous functions, polynomials

Continuity at a Point

Definition 4.5: Continuity

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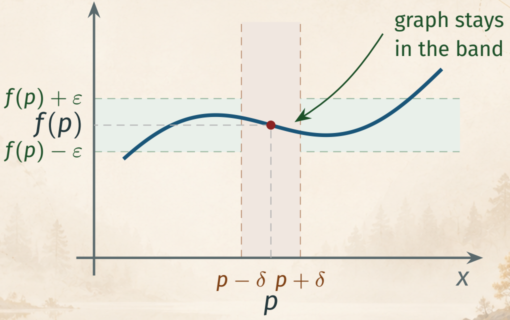
Suppose X and Y are metric spaces, $E \subset X$, $p \in E$, and $f : E \to Y$. Then f is continuous at p if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$d_Y(f(x), f(p)) < \varepsilon$$

for all points $x \in E$ for which $d_X(x, p) < \delta$.

If f is continuous at *every* point of E, then f is continuous on E.

Definition 4.5: The Picture



Key idea: the challenge–response game: given the ε-band, find a δ-strip whose graph piece stays inside it.

Two Observations

Observation 1: f must be defined at p

Continuity at p requires $p \in E$: the value $f(p)$ is part of the definition. (Contrast: for $\lim_{x \to p} f(x)$, we needed p to be a *limit point* of E, but $p \in E$ was not required.)

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Observation 2: isolated points are free

If p is an isolated point of E, then *every* function f on E is continuous at p: choose δ so small that the only $x \in E$ with $d_X(x, p) < \delta$ is $x = p$; then $d_Y(f(x), f(p)) = 0 < \varepsilon$.

Theorem 4.6: Continuity via Limits

Theorem 4.6

In the situation of Definition 4.5, assume also that p is a *limit point* of E. Then f is continuous at p if and only if

$$\lim_{x \to p} f(x) = f(p).$$

Proof

Compare Definitions 4.1 and 4.5. $\square$

Payoff: all the limit machinery of Section 4.1 (sequences, limit laws) now applies to continuity.

Composition of Functions

Theorem 4.7: Composition of Continuous Functions

Theorem 4.7

Suppose X, Y, Z are metric spaces, $E \subset X$, $f : E \to Y$, $g : f(E) \to Z$, and

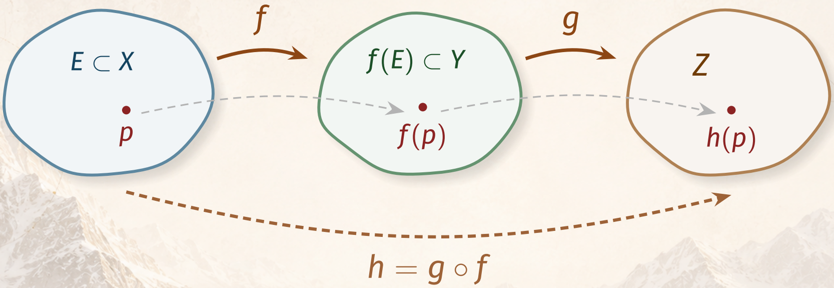
$$h(x) = g(f(x)) \qquad (x \in E).$$

If f is continuous at $p \in E$ and g is continuous at $f(p)$, then h is continuous at p.

$h = g \circ f$ is the composition of f and g.

Slogan: a continuous function of a continuous function is continuous.

Composition: The Picture



Proof strategy: chain the ε's — continuity of g produces an intermediate radius η, which becomes the "ε" challenge for f.

Proof of Theorem 4.7

Step 1: Use continuity of g at $f(p)$

Let $\varepsilon > 0$ be given. There exists $\eta > 0$ such that

$$d_Z(g(y), g(f(p))) < \varepsilon \quad \text{if } d_Y(y, f(p)) < \eta \text{ and } y \in f(E).$$

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There exists $\delta > 0$ such that

$$d_Y(f(x), f(p)) < \eta \quad \text{if } d_X(x, p) < \delta \text{ and } x \in E.$$

Hence, for such x:

$$\boxed{d_Z(h(x), h(p)) = d_Z(g(f(x)), g(f(p))) < \varepsilon.} \qquad \square$$

The Topological Characterization

Theorem 4.8: Continuity via Open Sets

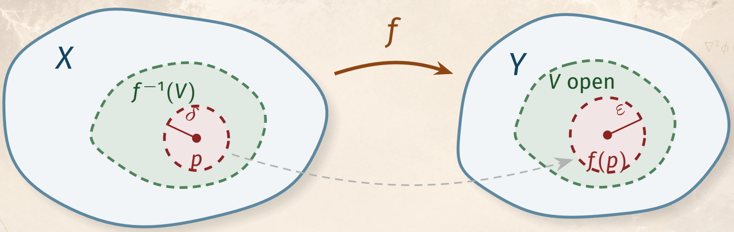
Theorem 4.8

A mapping f of a metric space X into a metric space Y is continuous on X if and only if $f^{-1}(V)$ is open in X for every open set V in Y.

Recall (Definition 2.2): the *inverse image* $f^{-1}(V) = \{x \in X : f(x) \in V\}$.

Why it matters: no ε, no δ, no distances — continuity becomes a statement about open sets alone.

Theorem 4.8: The Picture



Goal ($\Rightarrow$): every $p \in f^{-1}(V)$ is an *interior point* of $f^{-1}(V)$.

Proof of Theorem 4.8 ($\Rightarrow$)

Setup: continuity $\Rightarrow$ inverse images open

Suppose f is continuous on X and $V \subset Y$ is open. Let $p \in f^{-1}(V)$, so $f(p) \in V$.

- V open $\implies$ there is $\varepsilon > 0$: $y \in V$ whenever $d_Y(f(p), y) < \varepsilon$

Proof of Theorem 4.8 ($\Rightarrow$)

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Suppose f is continuous on X and $V \subset Y$ is open. Let $p \in f^{-1}(V)$, so $f(p) \in V$.

- V open $\implies$ there is $\varepsilon > 0$: $y \in V$ whenever $d_Y(f(p), y) < \varepsilon$
- f continuous at p $\implies$ there is $\delta > 0$: $d_Y(f(x), f(p)) < \varepsilon$ whenever $d_X(x, p) < \delta$

Hence $d_X(x, p) < \delta \implies f(x) \in V \implies x \in f^{-1}(V)$:

$$\boxed{p \text{ is an interior point of } f^{-1}(V).}$$

Proof of Theorem 4.8 ($\Leftarrow$)

Converse: open inverse images $\Rightarrow$ continuity

Suppose $f^{-1}(V)$ is open in X for *every* open $V \subset Y$. Fix $p \in X$ and $\varepsilon > 0$. Let

$$V = \{ y \in Y : d_Y(y, f(p)) < \varepsilon \}.$$

- V is open (an open ball) $\implies f^{-1}(V)$ is open, and $p \in f^{-1}(V)$

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$$V = \{\, y \in Y : d_Y(y, f(p)) < \varepsilon \,\}.$$

- V is open (an open ball) $\implies$ $f^{-1}(V)$ is open, and $p \in f^{-1}(V)$
- Hence there is $\delta > 0$: $x \in f^{-1}(V)$ whenever $d_X(p, x) < \delta$
- But $x \in f^{-1}(V) \Rightarrow f(x) \in V \Rightarrow d_Y(f(x), f(p)) < \varepsilon$ $\square$

Corollary: Continuity via Closed Sets

Corollary

A mapping f of a metric space X into a metric space Y is continuous if and only if $f^{-1}(C)$ is closed in X for every closed set C in Y.

Proof

A set is closed iff its complement is open, and inverse images respect complements:

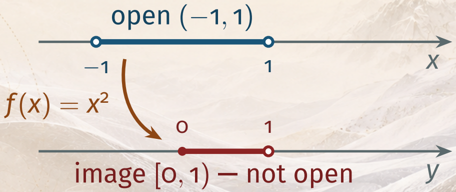
$$f^{-1}(E^c) = \left[f^{-1}(E)\right]^c \qquad (E \subset Y). \qquad \square$$

Caution: Inverse Images, Not Images!

A tempting — but false — misreading

Theorem 4.8 is about inverse images. A continuous f need *not* map open sets to open sets!

Counterexample:
$f(x) = x^2$ is continuous on R, but $f((-1,1)) = [0,1)$ is *not open* in R.



Algebra of Continuous Functions

Theorem 4.9: Sums, Products, Quotients

Theorem 4.9

Let f and g be complex continuous functions on a metric space X. Then

$$f + g, \qquad fg, \qquad f/g$$

are continuous on X (for f/g: assume $g(x) \neq 0$ for all $x \in X$).

Proof

At *isolated* points of X: nothing to prove (Observation 2). At *limit* points: combine Theorem 4.4 (limit laws) with Theorem 4.6. $\square$

Theorem 4.10: Vector-Valued Functions

Theorem 4.10 (a)

Let $f_1, \ldots, f_k$ be real functions on a metric space X, and let $\mathbf{f} : X \to R^k$ be defined by

$$\mathbf{f}(x) = (f_1(x), \ldots, f_k(x)) \qquad (x \in X); \tag{7}$$

then $\mathbf{f}$ is continuous if and only if each of $f_1, \ldots, f_k$ is continuous.

$f_1, \ldots, f_k$ are the components of $\mathbf{f}$.

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Theorem 4.10 (b)

If $\mathbf{f}, \mathbf{g} : X \to R^k$ are continuous, so are $\mathbf{f} + \mathbf{g}$ and $\mathbf{f} \cdot \mathbf{g}$.

Note: $\mathbf{f} + \mathbf{g}$ maps into R^k; the dot product $\mathbf{f} \cdot \mathbf{g}$ is a *real* function on X.

Proof of Theorem 4.10

Proof of (a): one inequality does it all

For $j = 1, \ldots, k$:

$$|f_j(x) - f_j(y)| \;\le\; |\mathbf{f}(x) - \mathbf{f}(y)| \;=\; \left\{ \sum_{i=1}^{k} |f_i(x) - f_i(y)|^2 \right\}^{1/2}$$

- $(\Rightarrow)$ $\mathbf{f}$ continuous: left inequality forces each f_j continuous (same δ works)
- $(\Leftarrow)$ all f_i continuous: right side small when each term is small

Proof of (b)

Follows from (a) and Theorem 4.9:
The components of $\mathbf{f} + \mathbf{g}$ are $f_i + g_i$, and $\mathbf{f} \cdot \mathbf{g} = \sum_i f_i g_i$. $\square$

Examples: A Library of Continuous Functions

Examples 4.11: Coordinate Functions

The coordinate functions ϕ_i

If $x_1, \ldots, x_k$ are the coordinates of $\mathbf{x} \in R^k$, the functions

$$\phi_i(\mathbf{x}) = x_i \qquad (\mathbf{x} \in R^k) \tag{8}$$

are continuous on R^k:

$$|\phi_i(\mathbf{x}) - \phi_i(\mathbf{y})| \le |\mathbf{x} - \mathbf{y}| \quad \Longrightarrow \quad \text{take } \delta = \varepsilon \text{ in Definition 4.5.}$$

Examples 4.11: Monomials and Polynomials

From coordinates to polynomials (via Theorem 4.9)

Every *monomial*

$$x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k} \tag{9}$$

($n_1, \ldots, n_k$ nonnegative integers) is continuous on R^k — repeated products of the ϕ_i. Constants are continuous, so every polynomial

$$P(\mathbf{x}) = \sum c_{n_1 \cdots n_k} x_1^{n_1} \cdots x_k^{n_k} \qquad (\mathbf{x} \in R^k) \tag{10}$$

(complex coefficients, finitely many terms) is continuous on R^k.

Examples 4.11: Rational Functions

Rational functions

Every rational function in $x_1, \ldots, x_k$ (a quotient of two polynomials of the form (10)) is continuous on R^k wherever the denominator is nonzero.

Why: polynomials are continuous + quotient rule of Theorem 4.9.

Example: $\dfrac{x_1 x_2 - 1}{x_1^2 + x_2^2}$ is continuous on $R^2 \setminus \{\mathbf{0}\}$.

Examples 4.11: The Norm is Continuous

The mapping $\mathbf{x} \to |\mathbf{x}|$

The *reverse* triangle inequality:

$$\big|\,|\mathbf{x}| - |\mathbf{y}|\,\big| \le |\mathbf{x} - \mathbf{y}| \qquad (\mathbf{x}, \mathbf{y} \in R^k), \tag{11}$$

so $\mathbf{x} \to |\mathbf{x}|$ is continuous on R^k (again $\delta = \varepsilon$).

Why: $|\mathbf{x}| = |(\mathbf{x} - \mathbf{y}) + \mathbf{y}| \le |\mathbf{x} - \mathbf{y}| + |\mathbf{y}| \implies |\mathbf{x}| - |\mathbf{y}| \le |\mathbf{x} - \mathbf{y}|$ (swap $\mathbf{x} \leftrightarrow \mathbf{y}$).

Combining with composition (Theorem 4.7)

If $\mathbf{f} : X \to R^k$ is continuous, then $\phi(p) = |\mathbf{f}(p)|$ is continuous real function on X.

A Simplifying Remark

Remark 4.12: Forget the Complement

Remark 4.12

Continuity was defined for $f : E \to Y$ with $E \subset X$ — but the complement of E plays no role in the definition. (The situation was *different* for limits of functions.)

Consequence: we may discard the ambient space and simply study continuous mappings of one metric space into another (E itself is a metric space!).

This simplifies statements and proofs — already used in Theorems 4.8–4.10, and again in the next section.

Summary

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Key takeaways from this section:

- Definition 4.5: f continuous at p: ε–δ control of $d_Y(f(x), f(p))$; automatic at isolated points
- Theorem 4.6: at limit points, continuity $\iff \lim_{x\to p} f(x) = f(p)$

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- Theorem 4.6: at limit points, continuity $\iff \lim_{x\to p} f(x) = f(p)$
- Theorem 4.7: compositions $g \circ f$ of continuous functions are continuous
- Theorem 4.8: f continuous $\iff f^{-1}(V)$ open for all open V ($\iff f^{-1}(C)$ closed for all closed C)

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- Theorem 4.7: compositions $g \circ f$ of continuous functions are continuous
- Theorem 4.8: f continuous $\iff f^{-1}(V)$ open for all open V ($\iff f^{-1}(C)$ closed for all closed C)
- Theorems 4.9–4.10 + Examples 4.11: sums, products, quotients, components; polynomials, rational functions, $|\mathbf{x}|$ all continuous

Coming up next: *Continuity and Compactness* — extreme values, uniform continuity.